

WASP-80b

R-2308

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-80_230630 · created 2026-08-17T23:39:27+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460125.9006 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1753 +/- 0.0079
Transit depth (Rp/Rs)^2	3.07 +/- 0.28 [%]
Orbital Inclination (inc)	89.95 +/- 0.4
Airmass coefficient 1 (a1)	1.216 +/- 0.011
Airmass coefficient 2 (a2)	-0.148 +/- 0.025
Scatter in the residuals of the lightcurve fit is	0.91 %
Best Comparison Star	None
Optimal Aperture	5.76
Optimal Annulus	7.55
Transit Duration (day)	0.09053 +/- 0.00077

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1753 $\pm$ 0.0079 vs 0.1714 (deviation 0.42 σ)
Duration fit vs published	0.09053 d vs 0.0892 d (deviation 1.5 σ)
Mid-time O–C	4.3 min
Depth SNR	18.8
Residual scatter	0.91 %

Light curve

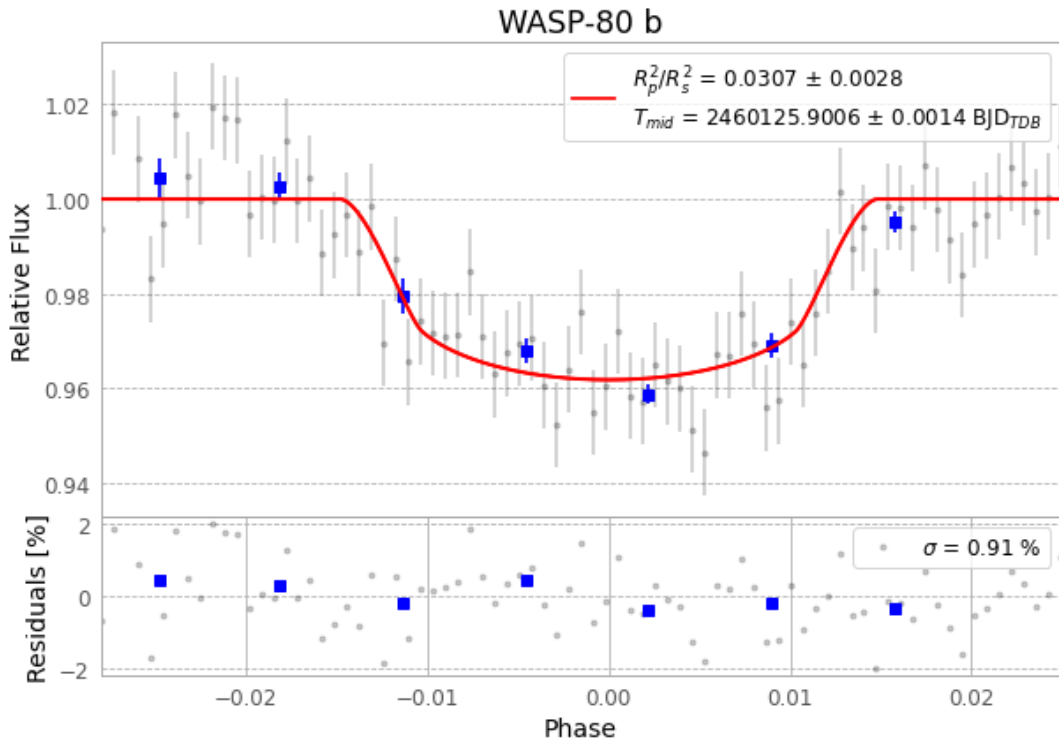


Figure 1 — FinalLightCurve_WASP-80 b_2023-06-30.png (SHA-256 5b75bcb35664a630...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [357, 187], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c0d2483f7bf5cb27...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

79 FITS frames (52 MB), WASP-80230630072415.FITS → WASP-80230630111816.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-80-230630.log (3b9188159e4b...), FinalLightCurve_WASP-80 b_2023-06-30.png (5b75bcb35664...), FinalLightCurve_WASP-80 b_2023-06-30.csv (c171e44014b4...)

Compute resources

Wall time 128.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-17 23:39Z.